

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and shade your choice on the answer sheet provided.

- 1 *Use of the Data Booklet is relevant to this question.*

How many electrons have to be removed to ionise 1.0×10^{-6} mol of Ne atoms to Ne^+ ions in a neon advertising tube?

- A** $\frac{6.02 \times 10^{23}}{1.0 \times 10^{-6}}$
- B** $1.0 \times 10^{-6} \times 6.02 \times 10^{23}$
- C** $\frac{1.0 \times 10^{-6} \times 6.02 \times 10^{23}}{20.2}$
- D** $\frac{1.0 \times 10^{-6} \times 6.02 \times 10^{23}}{9.65 \times 10^4}$

- 2 Which factor helps to explain why the first ionisation energies of the Group I elements decrease from lithium to sodium to potassium to rubidium?

- A** The nuclear charge of the elements increases.
- B** The outer electrons are in an 's' subshell.
- C** The repulsion between spin-paired electrons increases.
- D** The shielding effect of the inner shells increases.

- 3 A sample of m g of an organic compound is vaporised in a gas syringe and occupies $V \text{ cm}^3$ at $T \text{ K}$ and $p \text{ atm}$.

What is the relative molecular mass, M_r , of the compound?

- A** $M_r = \frac{m \times 22400 \times T}{p \times V \times 273}$
- B** $M_r = \frac{m \times 22400 \times (T + 273)}{p \times V \times 273}$
- C** $M_r = \frac{m \times 22400 \times 273 \times p}{V \times T}$
- D** $M_r = \frac{m \times 22400 \times 273 \times p}{V \times (T + 273)}$

- 4 The application of pressure can cause a gas to liquefy because the gas molecules are pushed together such that intermolecular forces become significant. The molecules can no longer move freely and the gas condenses into a liquid.

Gases can only be liquefied by pressure alone if their temperature is below their 'critical temperature', T_c . Above the critical temperature, no amount of pressure will liquefy a gas.

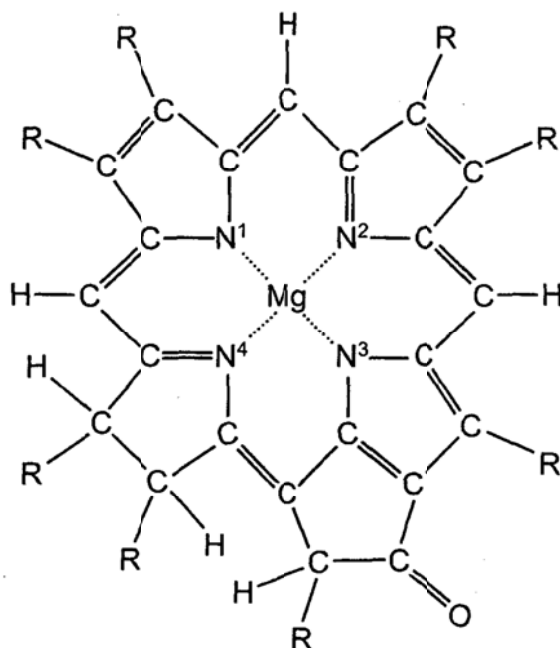
Methane gas, ammonia gas and steam have similar relative molecular mass.

What is the order of their critical temperatures?

	highest	—————→	lowest
A	methane	ammonia	steam
B	methane	steam	ammonia
C	ammonia	steam	methane
D	steam	ammonia	methane

- 5 Which type of structure, present in ceramics, explains why these materials are strong?
- A** ionic
- B** layer
- C** giant molecular
- D** metallic

- 6 Plants appear green due to the presence of chlorophyll. There are several closely related chlorophylls and the diagram shows a simplified version of one. The various different side-groups are all shown as R.

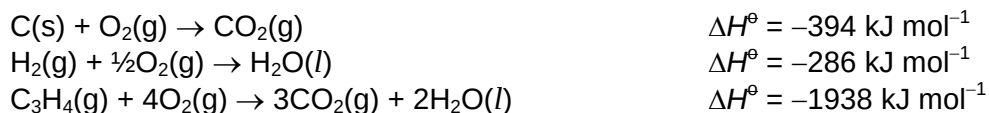


Note that the four N atoms and the Mg ion are planar.

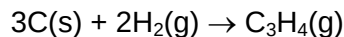
Which of the descriptions of the bonds between Mg and the numbered N atoms is most likely to be correct?

	N atoms numbered	
	1 and 3	2 and 4
A	co-ordinate	ionic
B	co-ordinate	π
C	ionic	co-ordinate
D	π	co-ordinate

- 7 The standard enthalpy changes for three reactions are given by the equations:



What is the standard enthalpy change for the following reaction?



- A $-1258 \text{ kJ mol}^{-1}$
 B $+1258 \text{ kJ mol}^{-1}$
 C -184 kJ mol^{-1}
 D $+184 \text{ kJ mol}^{-1}$
- 8 Sulfur exists in two allotropic forms, monoclinic sulfur and rhombic sulfur. At temperatures below 93°C , monoclinic sulfur changes into rhombic sulfur. Data for each form of sulfur are given in the table.

	$\Delta H_f^\circ / \text{kJ mol}^{-1}$	$S_f^\circ / \text{J K}^{-1} \text{mol}^{-1}$
monoclinic	+0.30	32.6
rhombic	0	31.7

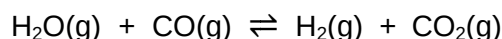
What is the expression for ΔG° , in J mol^{-1} , for the formation of the rhombic sulfur from monoclinic sulfur at 90°C ?

- A $\Delta G^\circ = -0.30 - 363(-0.9)$
 B $\Delta G^\circ = -0.30 - 90(+0.9)$
 C $\Delta G^\circ = -300 - 90(+0.9)$
 D $\Delta G^\circ = -300 - 363(-0.9)$
- 9 Each of the following equilibria is subjected to two changes done separately.
- pressure is reduced at constant temperature
 - temperature is increased at constant pressure

For which of the following equilibria will both of these changes result in an increase in the proportion of products?

- A $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$ $\Delta H > 0$
 B $2\text{NO(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{NO}_2\text{(g)}$ $\Delta H < 0$
 C $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$ $\Delta H > 0$
 D $4\text{NH}_3\text{(g)} + 5\text{O}_2\text{(g)} \rightleftharpoons 4\text{NO(g)} + 6\text{H}_2\text{O(g)}$ $\Delta H < 0$

- 10 A mixture of steam and carbon monoxide will form an equilibrium mixture as shown by the equation.

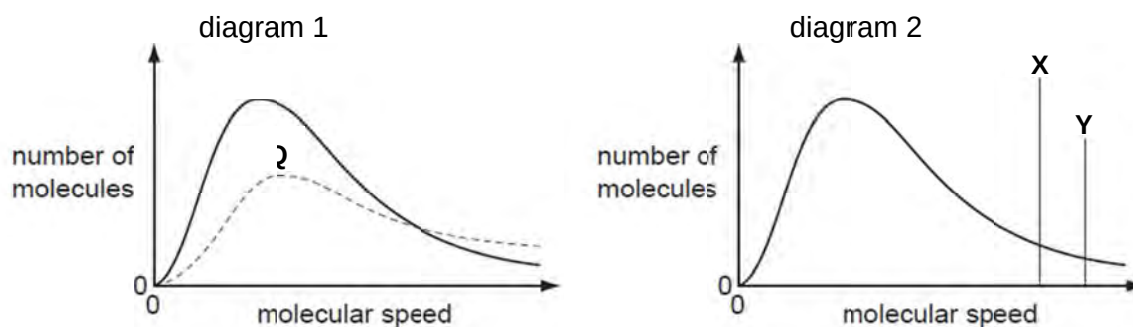


If 1.0 mol of steam and 1.0 mol of carbon monoxide are allowed to reach equilibrium, 33.3% of the resulting mixture is carbon dioxide.

What is the value of the equilibrium constant, K_p ?

- A 4.0 B 2.0 C 1.0 D 0.25

- 11 Different Boltzmann distributions are shown in the diagrams.



In diagram 1, one curve **P** or **Q** corresponds to a temperature higher than that of the other curve.

In diagram 2, one line **X** or **Y** corresponds to the activation energy for a catalysed reaction and the other line corresponds to the activation energy of the same reaction when uncatalysed.

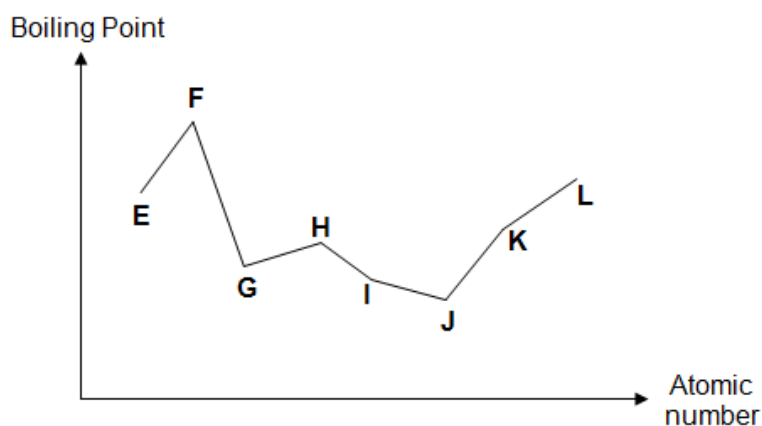
Which combination gives the correct curve and line?

	higher temperature	presence of catalyst
A	P	X
B	P	Y
C	Q	X
D	Q	Y

- 12 Although beryllium and aluminium are in different groups in the Periodic Table, they exhibit similar chemical properties due to the *diagonal relationship*.

Which property of beryllium compounds is **not** likely to be correct?

- A The charge density of Be^{2+} is similar to that of Al^{3+} .
- B Beryllium chloride dissolves in water to form a neutral solution.
- C Beryllium chloride exists as a simple molecular compound.
- D Beryllium chloride can form a dimer.
- 13 The graph below shows the variation in the boiling points for eight consecutive elements in the Periodic Table, all with atomic numbers from 10 to 20. The letters do not represent any element in the Periodic Table.



Which of the following can be deduced from the graph above?

- A Element E and boron are in the same group.
- B Element K does not conduct electricity.
- C Element G can exist in two allotropic forms.
- D Element H has a smaller atomic radius than I.
- 14 Which compound is **not** a product of the reaction between an oxide of a third period element and water?

- A NaOH B H_3PO_4 C H_2SO_4 D H_2SiO_3

- 15** In the gravimetric analysis of a mixture of Group II salts, an excess of dilute hydrochloric acid was added. A colourless gas was evolved together with a clear solution.

Which of the following pair of compounds could **not** be present in the mixture?

- A** MgCO_3 and MgSO_4
- B** MgSO_4 and CaCO_3
- C** CaCO_3 and $\text{Ba}(\text{NO}_3)_2$
- D** CaCO_3 and BaSO_4
- 16** Which of the following **decreases** down Group II?
- A** magnitude of the enthalpy change of hydration of the cations
- B** thermal decomposition temperature of the nitrates
- C** pH of the oxides in water
- D** reducing power of the elements
- 17** The number of moles of chlorine that react with 1 mole of **U** is twice the number of moles of chlorine that react with 1 mole of **V**.

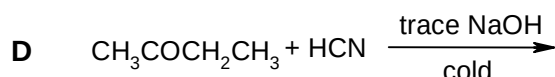
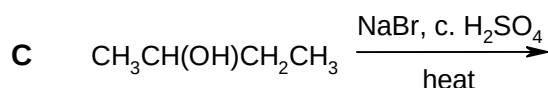
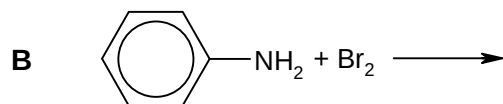
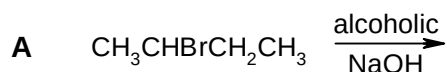
3 students proposed the following identities of **U** and **V**.

	U	V
student 1	Mg(s)	Na(s)
student 2	$\text{H}_2(\text{g})$	KBr(aq)
student 3	cold NaOH(aq)	hot NaOH(aq)

Which of the following statements is true?

- A** Only student 1 is correct.
- B** Only student 1 & 2 are correct.
- C** Only student 2 & 3 are correct.
- D** Only student 3 is correct.

- 18 In which of the following reactions will the affected carbon atom change from being sp^2 hybridised to sp^3 hybridised?



- 19 What is the total possible number of structural and stereoisomers of the formula $\text{C}_3\text{H}_5\text{Br}$ that are non-cyclic?

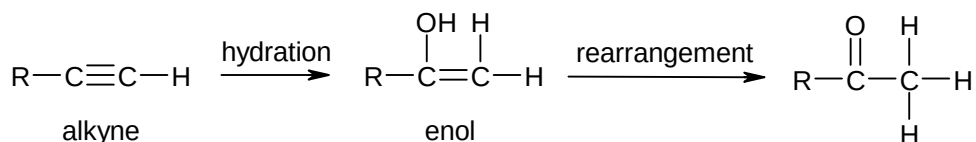
A 2

B 3

C 4

D 5

- 20 An alkyne undergoes hydration in a similar mechanism as an alkene. However, the formation of an enol is unstable and would undergo rearrangement to form a carbonyl compound as shown below.



Which of the following product will not be formed when the corresponding alkyne undergoes hydration?

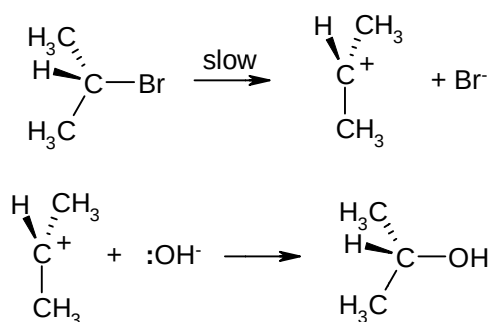
	alkyne	product
A	$\text{H}-\text{C}\equiv\text{C}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{CH}_3 \end{array}$
B	$\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C}-\text{CH}_2\text{CH}_3 \end{array}$
C	$\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{H}$	$\begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \end{array}$
D	$\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{CH}_3$	$\begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{CH}_3 \end{array}$

10

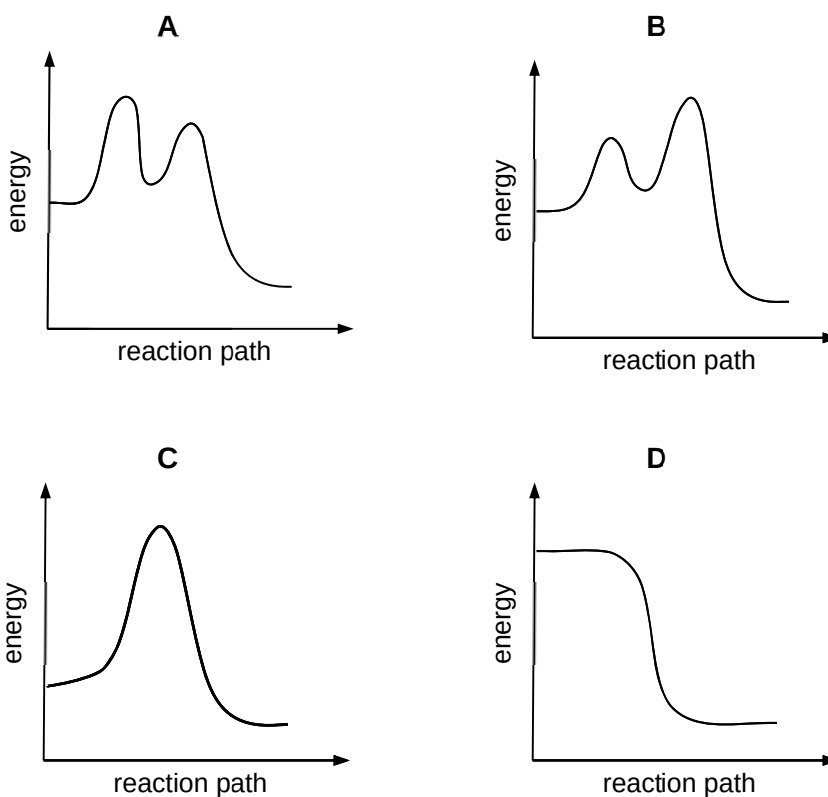
- 21 It has been estimated that for every atom of chlorine generated from a fluorohalogenoalkane in the stratosphere, one hundred thousand molecules of ozone may be destroyed.

Which statement about chlorine atoms gives an explanation of this?

- A They achieve high activation energies by absorbing ultraviolet radiation.
- B They are regenerated in a chain propagation reaction with ozone.
- C They remove oxygen from the ozone-oxygen equilibrium.
- D They initiate the ionisation of oxygen molecules.
- 22 A possible mechanism for the hydrolysis of 2-bromopropane is shown below.



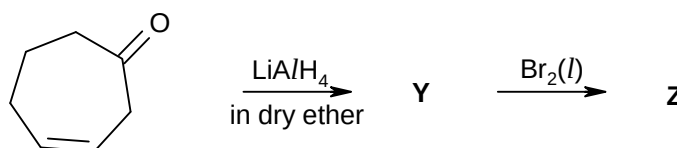
Which diagram represents the reaction profile for this reaction?



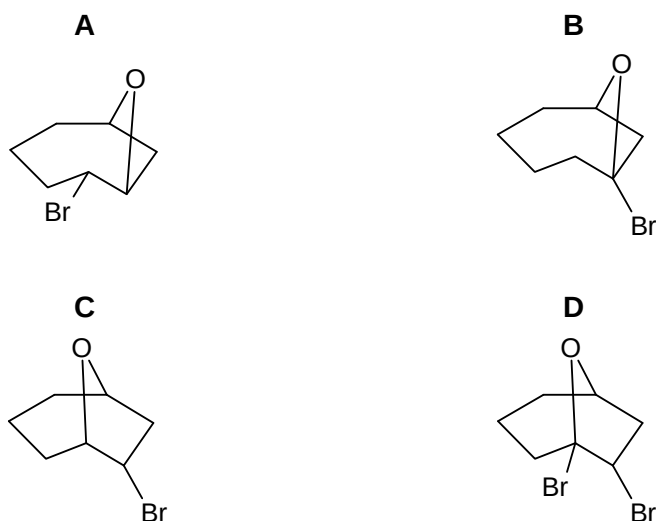
- 23 A mixture is made by adding together one mole of sodium hydroxide, one mole of phenol, $\text{C}_6\text{H}_5\text{OH}$, and one mole of hexan-1-ol, $\text{C}_6\text{H}_{13}\text{OH}$, each solute being in aqueous solution.

Which organic species are present in this final solution?

- A mainly $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_{13}\text{OH}$
 B mainly $\text{C}_6\text{H}_{13}\text{OH}$ and $\text{C}_6\text{H}_5\text{O}^-\text{Na}^+$
 C mainly $\text{C}_6\text{H}_5\text{OH}$ and $\text{C}_6\text{H}_{13}\text{O}^-\text{Na}^+$
 D approximately equal quantities of $\text{C}_6\text{H}_5\text{OH}$, $\text{C}_6\text{H}_5\text{O}^-\text{Na}^+$, $\text{C}_6\text{H}_{13}\text{OH}$ and $\text{C}_6\text{H}_{13}\text{O}^-\text{Na}^+$
- 24 The reaction scheme below gives product **Z**.



Which of the following is the most likely structure for **Z**?



- 25 The table shows the results of simple tests on a compound **Q**.

reagents	observations
2,4-dinitrophenylhydrazine	orange ppt
Tollen's reagent	no ppt
aqueous alkaline iodine	yellow ppt

From the results of the tests, what could **Q** be?

- A CH_3CHO
 B $\text{CH}_3\text{CH}_2\text{CHO}$
 C $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
 D CH_3COCH_3

12

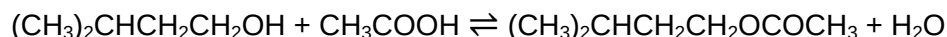
- 26 Calcium ethanoate produces propanone when heated as shown in the equation below.



When this reaction is carried out by heating a 1 : 1 molar mixture of calcium propanoate, $(\text{CH}_3\text{CH}_2\text{CO}_2)_2\text{Ca}$, and calcium methanoate, $(\text{HCO}_2)_2\text{Ca}$, a mixture of three carbonyl compounds is produced.

Which of the following carbonyl compounds is **not** formed?

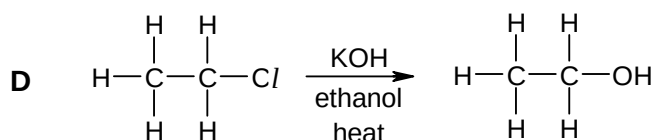
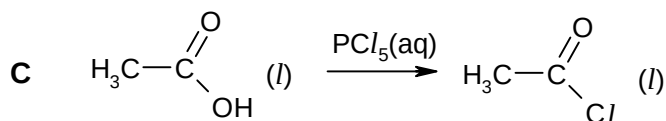
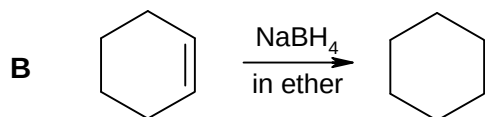
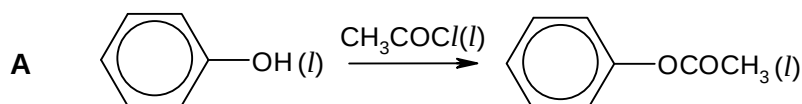
- A HCHO
 B $\text{CH}_3\text{CH}_2\text{CHO}$
 C $\text{CH}_3\text{CH}_2\text{COCH}_3$
 D $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- 27 Isopentyl acetate, $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$, is a colourless liquid used as banana flavouring in fruit sweets. It can be prepared by reacting an excess of ethanoic acid with isopentyl alcohol in the presence of concentrated sulfuric acid.



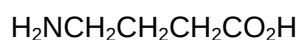
Which of the following steps are involved in the purification of the product?

- I. adding anhydrous calcium chloride
 II. distillation
 III. separation of liquids using a separating funnel
 IV. shaking the mixture with NaHCO_3

- A I, II, III B II, III, IV C I, II, IV D I, II, III, IV
- 28 Which of the following shows the correct reagents and conditions to produce the desired product?



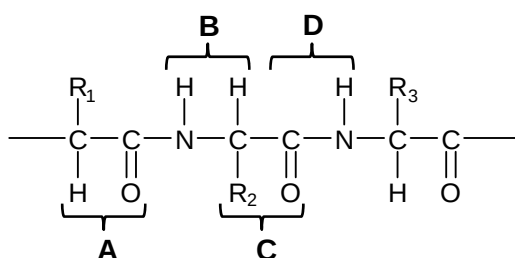
- 29 GABA is a neuro-transmitter released by red algae which encourages shellfish larvae to settle on the ocean bed.



GABA

In what way does GABA differ from amino acids obtained by the hydrolysis of protein?

- A It does not form a zwitterion.
- B It is not soluble in water.
- C It is not a 2-aminocarboxylic acid.
- D It does not form a polyamide linkage.
- 30 A form of blindness is curable by replacing the lens of the eye. The lens is held in place by proteins. Part of the protein structure is shown below.



Before the lens is removed, the protein structure is weakened by injecting an enzyme into the eye which hydrolyses some of the peptide linkages.

Which labelled part of the protein structure will be hydrolysed by the enzyme?

14

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to pick a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 Which statements about relative molecular mass are correct?

- 1** It is the sum of the relative atomic masses of all the atoms within the molecule.
- 2** It is the ratio of the average mass of a molecule to the mass of a ^{12}C atom.
- 3** It is the ratio of the mass of 1 mol of molecules to the mass of 1 mol of ^1H atoms.

32 In which of the following conversions does the oxidation number of nitrogen change by two?

- 1** $\text{NH}_2\text{OH} \rightarrow \text{NH}_3$
- 2** $\text{N}_2 \rightarrow \text{NO}$
- 3** $\text{NO}_2 \rightarrow \text{HNO}_3$

- 33** The graph shows the results of an investigation of the initial rate of hydrolysis maltose by the enzyme amylase. In the experiments, the initial concentration of maltose was varied but that of amylase was kept constant.



Which conclusions can be deduced from these results?

- 1 When [maltose] is low, the rate is first order with respect to [maltose].
 - 2 When [maltose] is high, the rate is independent of [maltose].
 - 3 When [maltose] is high, the rate is independent of [amylase].
- 34** Which set of solutions below gives a buffer solution when they are mixed?
- 1 50 cm³ of 0.1 mol dm⁻³ CH₃COOH and 50 cm³ of 0.1 mol dm⁻³ CH₃COONa
 - 2 50 cm³ of 0.1 mol dm⁻³ CH₃COOH and 25 cm³ of 0.1 mol dm⁻³ NaOH
 - 3 50 cm³ of 0.1 mol dm⁻³ C₆H₅COONa and 25 cm³ of 0.1 mol dm⁻³ HCl
- 35** Hydroxyapatite, a basic calcium phosphorus, Ca(OH)₂·3Ca₃(PO₄)₂, is the mineral found in bone. In older people, calcium ions can be lost from the hydroxyapatite, weakening the bone structure. In such cases, strontium salts are administered to strengthen the bone. The strontium ions replace the lost calcium ions in the hydroxyapatite.
- [numerical value of K_{sp} : Ca(OH)₂, 5.2×10^{-6} ; Sr(OH)₂, 1.46×10^{-1}]
- Which statements are correct?
- 1 Strontium ions are nearly the same size as calcium ions and so may easily replace them in the hydroxyapatite.
 - 2 Strontium hydroxide is less soluble than calcium hydroxide and so will precipitate better in the bone structure.
 - 3 There is ionic, covalent and metallic bonding in hydroxyapatite which gives it strength.

16

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

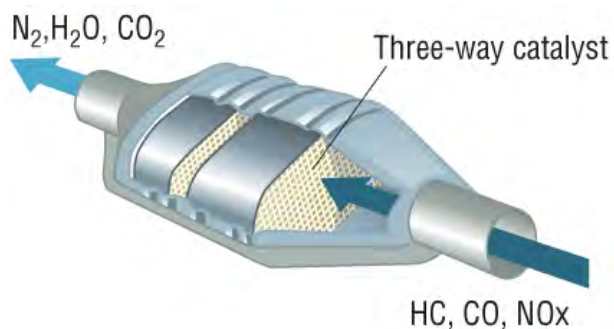
No other combination of statements is used as a correct response.

- 36** Zirconium has an atomic number of 40.

Which zirconium compounds are likely to exist?

- 1** Zr_2O_7
- 2** ZrOF_2
- 3** $[\text{Zr}(\text{NH}_3)_6]\text{Cl}_3$

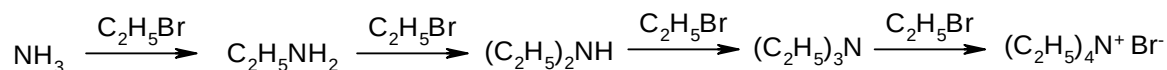
- 37** The diagram shows the structure of a catalytic converter as fitted into the exhaust system of a car. Harmful gases are converted into carbon dioxide, nitrogen and water vapour.



Which reactions between the stated compounds could take place on the surface of the catalyst?

- 1** hydrocarbons + oxides of nitrogen → carbon dioxide + water + nitrogen
- 2** carbon monoxide + oxides of nitrogen → carbon dioxide + nitrogen
- 3** carbon monoxide + hydrocarbon → carbon dioxide + water

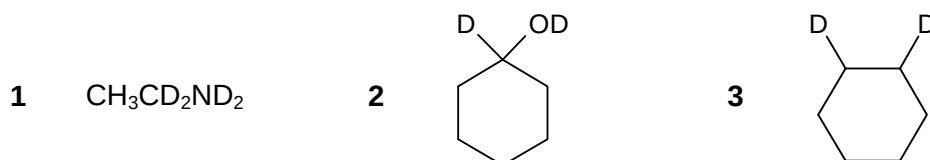
- 38** Ammonia reacts with excess bromoethane when heated in alcoholic medium to form a quaternary ammonium salt in four steps.



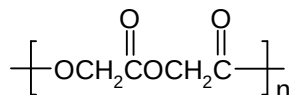
Which of the following statements about the reaction is correct?

- 1** In each step, the attacking nucleophile is a stronger nucleophile than that in the previous step.
 - 2** In each step, the reaction is slower than the previous step due to increased steric hindrance.
 - 3** In each step, an intermediate with a sp^2 hybridised carbon atom is formed.
- 39** Deuterium, D, is an isotope of hydrogen, ${}^2_1\text{H}$.

Which compound can be formed by the addition of D_2 to another molecule, in the presence of a platinum catalyst?



- 40** Dexon is a synthetic polymer that is routinely used to stitch wounds together during surgery. It is broken down into smaller molecules over three to six months. Patients, therefore do not need to undergo a second surgery to remove the stitches.



Dexon

Which statements explain Dexon's suitability for this function?

- 1** Dexon can be hydrolysed in the body.
- 2** Dexon is broken down slowly due to its large size.
- 3** The small molecules produced are water soluble.

~ END OF PAPER ~